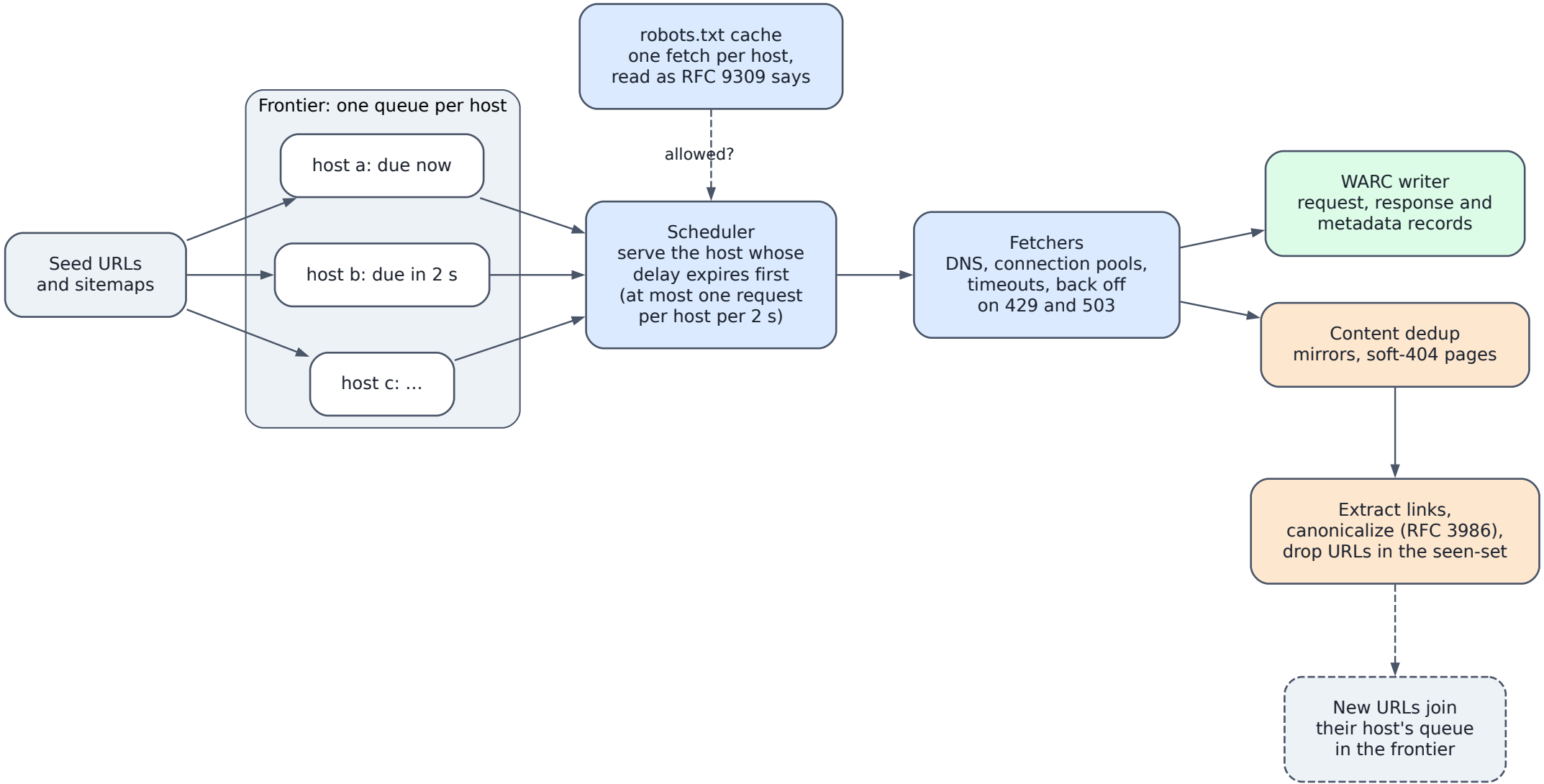
Seed URLs
and sitemaps
Frontier: one queue per host
host a: due now
host b: due in 2 s
host c: ...
robots.txt cache
one fetch per host,
read as RFC 9309 says
allowed?
Scheduler
serve the host whose
delay expires first
(at most one request
per host per 2 s)
Fetchers
DNS, connection pools,
timeouts, back off
on 429 and 503
WARC writer
request, response and
metadata records
Content dedup
mirrors, soft-404 pages
Extract links,
canonicalize (RFC 3986),
drop URLs in the seen-set
New URLs join
their host's queue
in the frontier